

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.1	10	0.4	0.2	0.1	59.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.2	9.3	0.1	0.2	0	59.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.8	10	0.7	0.1	0.1	59.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.2	8.2	0.1	0.1	0.1	59.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.6	10.2	0.5	0.1	0.1	59.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.4	8.1	0.1	0.1	0.1	59.2
U2 - Efficacy additive alpha prior Beta(1, 1)	30	9.4	0.7	0.1	0.1	59.7
U3 - Efficacy additive alpha prior Beta(1, 1)	32.4	7.7	0.1	0.1	0.1	59.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	3.9	1.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	3.8	1.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	3.9	1.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	3.8	1.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.8	3.8	1.7	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.8	3.8	1.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.8	3.8	1.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.8	3.8	1.8	0.6	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2	0.5	0.3	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.2	19.3	31.9	9.5	2.2	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	39.3	19	29.9	8.4	1.6	1.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.9	19.4	31.8	9.7	2.4	1.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	39.6	18.3	29.9	8.6	1.8	1.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.9	18.1	33.3	8.6	2.3	1.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	40.6	17.1	30.2	8.4	1.9	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	36.4	17.3	33.2	8.7	2.6	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	40.9	17.2	30.3	7.7	2.1	1.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	5.4	7.3	3.7	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	5.5	7.2	3.7	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	5.5	7.3	3.6	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	5.4	7.2	3.6	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	5.3	7.3	3.6	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	5.4	7.3	3.6	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	5.4	7.3	3.6	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.9	5.4	7.2	3.7	1.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.5	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.5	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.4	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.4	1.3	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.4	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.4	1.3	0.8	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	43.8	19.3	6.8	1.5	10.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.8	47	16.5	5.2	1.3	10.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	45.1	18.9	6.6	1.6	10.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	46.3	17.1	4.9	1.2	10.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.2	41.9	20.5	7.1	2	10.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	43.7	19.2	4.7	1.2	10.2
U2 - Efficacy additive alpha prior Beta(1, 1)	18.2	42.6	20.4	6.7	1.9	10.2
U3 - Efficacy additive alpha prior Beta(1, 1)	20.5	43.8	18.7	5.4	1.3	10.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	7.9	5.9	3.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	7.9	6	3.4	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	7.8	6	3.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	7.9	6	3.4	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	7.7	6	3.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	7.6	6.1	3.4	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.2	7.7	6.1	3.4	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	7.6	6.1	3.4	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.2	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.2	0.9	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.9	8.5	23.2	41.6	16.8	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	9.6	20.9	40.6	14.1	1.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	9.1	23.5	41.6	16.4	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	10.1	19.6	40.7	15	1.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	9.7	23	41.1	16.9	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12	10.5	20.8	39.9	14.9	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	6.9	9.2	23.6	40.9	17.5	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	12.4	9.7	20.3	40.9	14.8	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	4.4	6.8	8	5.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	4.4	6.8	8.1	5.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	4.4	6.9	8	5.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	4.4	6.8	8.1	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	4.4	6.8	8	5.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	4.4	6.8	8	5.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	4.4	6.9	7.9	5.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	4.4	6.9	7.9	5.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.4	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.4	2.2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.5	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.5	2.2	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	34.9	26.3	20.4	9.5	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.5	41.5	24.6	15.6	6.6	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	33.4	27.7	19.6	10.1	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	40	24.4	16.4	6.1	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.6	31.1	28.8	20.3	10	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	37.2	27.1	15.6	6.4	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	5.7	33.2	27.7	18.3	10.9	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	36.6	27.5	15.1	6.9	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	6.5	6.9	5.8	5.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	6.6	6.7	5.9	5.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	6.5	6.8	5.8	5.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	6.6	6.7	5.8	5.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	6.5	6.8	5.7	5.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	6.4	6.8	5.8	5.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	6.5	6.8	5.7	5.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	6.5	6.8	5.7	5.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	1.7	2.1	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	1.6	2.1	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.6	2.1	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.7	2.1	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.7	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	1.7	2.1	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.3	12.7	2.8	0.8	0.3	14.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.1	12.3	2.4	0.8	0.2	14.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.1	12.4	2.9	1	0.3	14.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	72.3	11.1	1.5	0.6	0.2	14.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.3	11.3	3.1	1	0.2	14.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.3	10.6	2.2	0.6	0.2	14.1
U2 - Efficacy additive alpha prior Beta(1, 1)	71	10.9	2.9	0.9	0.2	14.1
U3 - Efficacy additive alpha prior Beta(1, 1)	72.9	10.1	2.1	0.7	0.2	14
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	4	2.1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	4	2.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.2	4	2.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.2	4	2.2	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.3	3.8	2.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	3.9	2.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.3	3.9	2.2	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.3	3.8	2.2	0.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.4	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.4	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.4	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.2	38.5	20.1	9.8	3.1	2.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.5	42	18	7.6	1.7	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.4	38	21.3	9.4	2.7	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.3	40.3	18.8	7.7	1.7	2.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27	36.6	21.9	9.8	2.5	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.5	39.5	19.5	7.3	2	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	27.4	37.3	21.4	9.1	2.6	2.2
U3 - Efficacy additive alpha prior Beta(1, 1)	29.1	39.3	19.7	7.5	2.1	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	7.3	5.6	3.6	2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	7.3	5.6	3.6	2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	7.2	5.7	3.6	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	7.2	5.7	3.6	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	7.1	5.7	3.6	2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	7	5.8	3.6	2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	7.1	5.7	3.7	2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	7	5.8	3.5	2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1	0.9	0.3	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	43.2	8.4	1.7	0.6	0.1	46
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.5	8.2	1.1	0.4	0.1	45.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	43.6	8.3	1.8	0.5	0.2	45.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	44.6	7.6	1.5	0.3	0.2	45.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	44	8	1.5	0.5	0.2	45.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.8	7.6	1.1	0.5	0.1	45.9
U2 - Efficacy additive alpha prior Beta(1, 1)	43.9	7.8	1.7	0.6	0.3	45.7
U3 - Efficacy additive alpha prior Beta(1, 1)	45.1	7.7	0.8	0.5	0.2	45.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	3.5	1.6	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	3.6	1.5	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	3.5	1.6	0.5	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	3.5	1.6	0.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	3.4	1.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	3.4	1.6	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.1	3.4	1.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.1	3.4	1.6	0.4	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.3	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	12.7	17.1	13.6	21.2	19.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.7	13.4	14.9	12.4	18.2	19.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.7	13.9	16.9	13.3	20.9	19.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	12.8	14.8	11.7	18.4	19.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	12.6	17.8	13.6	21.2	19.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.5	12.1	15.1	11.3	19.3	19.7
U2 - Efficacy additive alpha prior Beta(1, 1)	15	12.7	17.6	13.5	21.5	19.7
U3 - Efficacy additive alpha prior Beta(1, 1)	22.1	11.6	15.3	11.7	19.6	19.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	4.6	5.5	3.8	4.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	4.7	5.4	3.9	4.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	4.6	5.5	3.8	4.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	4.6	5.5	3.8	4.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	4.6	5.5	3.8	4.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	4.6	5.5	3.8	4.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	4.6	5.5	3.8	4.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	4.5	5.6	3.7	4.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.3	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.7	1.3	1.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.7	1.3	1.3	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.7	1.3	1.3	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.7	1.3	1.3	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.6	31	11.1	3.7	1.5	40.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	32.9	9	2.8	1.2	40.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.6	31.1	11	3.6	1.5	40.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	31.3	10.2	2.8	1.4	40.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	30.3	11.9	3.5	1.7	40.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.6	31.7	10.3	2.8	1.3	40.3
U2 - Efficacy additive alpha prior Beta(1, 1)	12.6	30.7	11.5	3.4	1.5	40.3
U3 - Efficacy additive alpha prior Beta(1, 1)	13.8	31.2	10.8	2.8	1.3	40.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	5.9	3.8	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.5	6	3.6	1.9	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	5.8	3.9	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.5	5.8	3.8	1.8	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	5.8	3.8	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	5.8	3.9	1.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.6	5.8	3.8	1.8	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.6	5.8	3.8	1.8	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.8	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.7	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.7	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.7	0.5	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.3	23.3	2.7	0.1	0	7.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.5	21.7	1.5	0.2	0	7.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.9	21.7	2.8	0.2	0	7.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.8	20.2	1.6	0.1	0	7.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70	19.6	2.9	0.1	0	7.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.1	17.7	1.8	0.2	0	7.2
U2 - Efficacy additive alpha prior Beta(1, 1)	69.6	20.2	2.4	0.4	0	7.4
U3 - Efficacy additive alpha prior Beta(1, 1)	73.1	17.8	1.6	0.1	0	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	5.9	3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	5.9	2.9	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.2	5.8	3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.2	5.9	3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	5.7	3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	5.7	2.9	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.3	5.8	3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.4	5.7	3	0.8	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.4	0.7	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.4	0.7	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15	14.4	11.3	36.6	19.5	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22	14.8	8.8	33.6	17.6	3.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.5	13.3	12	37	20	3.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.3	13.5	9.3	33.1	18.6	3.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	14.2	11.2	35.9	20.5	3.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	13.7	7.6	34.4	18.8	3.2
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	14.9	11.7	35.8	20.6	3.2
U3 - Efficacy additive alpha prior Beta(1, 1)	21.8	14.4	7.8	34	18.8	3.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.9	5.6	7.4	6.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.8	5.6	7.3	6.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	4.8	5.5	7.3	6.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	4.8	5.6	7.3	6.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.8	5.6	7.2	6.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.8	5.6	7.3	6.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.2	4.8	5.5	7.3	6.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	4.8	5.5	7.4	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	33.2	12	3.5	1.3	35.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	33.3	10.8	3.2	1.1	35.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.3	33.2	12.6	3.7	1.2	35
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.4	33.5	10.2	3.2	1.3	35.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.8	32.9	13.5	3.1	1.3	35.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.2	32.8	10.8	3.7	1.1	35.4
U2 - Efficacy additive alpha prior Beta(1, 1)	13.8	32.2	14.1	3.7	1.1	35.1
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	32.6	11.1	3.5	1.1	35.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	6.3	4	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.3	6.2	4.1	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.2	6.3	4.1	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.2	6.2	4.1	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.3	6.2	4.1	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.3	6.1	4.1	2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.3	6.1	4.1	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.3	6.1	4.1	2	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.9	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.9	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.6	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	17.6	13.7	26.9	12.1	7.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.6	18.8	10.2	23.4	8.6	7.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.7	19	14.2	25.5	12.1	7.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.8	18.1	9.5	22.8	9.3	7.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.8	17.7	15.1	25.2	12.7	7.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.6	17.3	10.9	21.9	9.9	7.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.8	17.4	14.9	25.9	12.6	7.4
U3 - Efficacy additive alpha prior Beta(1, 1)	32.1	18.2	10.4	21.4	10.5	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	5.6	5.9	5.8	4.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	5.5	5.9	5.8	4.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	5.6	6	5.6	4.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.1	5.6	6	5.6	4.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	5.6	6	5.7	4.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	5.6	5.9	5.7	4.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	5.5	6.1	5.6	4.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.1	5.6	6	5.7	4.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.5	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.5	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.3	1.5	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.3	1.4	1.9	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.5	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	1.4	2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.3	1.5	1.9	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.5	1.9	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	51.4	27.9	9.6	3.1	0.6	7.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	60.7	25	5.3	1.6	0.2	7.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	53.2	27.1	8.5	3.4	0.7	7.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	61.9	24.4	4.6	1.8	0.2	7.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.8	27.2	9.3	3	0.5	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	65	20.9	5.2	1.6	0.2	7.1
U2 - Efficacy additive alpha prior Beta(1, 1)	52.7	27.6	8.9	3.1	0.5	7.2
U3 - Efficacy additive alpha prior Beta(1, 1)	63.9	22.2	5.2	1.4	0.2	7.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	7.3	5.6	3	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	7.4	5.7	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.8	7.3	5.8	3	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	7.3	5.7	3	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	7.3	5.7	3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12	7.2	5.8	3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	7.3	5.8	3	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12	7.2	5.7	3	1.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.5	1.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.5	1.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.6	1.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.6	1.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.6	1.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.5	1.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.6	1.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.6	1.5	1.2	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31	25.4	16.3	16.7	10.5	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.4	29.7	14	12.2	5.6	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.7	24.2	17.7	16.5	10.8	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	39.4	27.8	15.1	12.6	5	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.7	23.2	16.2	17.9	10.9	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	39	27.8	15.8	12.4	4.9	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	31.3	23.9	16.6	17.7	10.4	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	39.9	27	15.6	12.6	4.8	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	5.7	5.5	5.9	7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.9	5.7	5.4	5.9	7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.7	5.5	5.8	7.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	5.6	5.4	5.9	7.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	5.5	5.4	5.8	7.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	5.6	5.4	5.8	7.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	5.6	5.5	5.8	7.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.9	5.5	5.4	5.9	7.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.2	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.2	1.8	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	1.7	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.1	1.8	2.6	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	1.8	2.6	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.2	53.2	16.1	1.2	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.3	54.1	13.3	1	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.5	50.3	17.6	1.3	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.9	50.7	15	1.1	0	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.8	49.8	16.7	1.4	0	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.6	48.8	14.1	1.2	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	31.2	49.9	17.1	1.5	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	35	49.1	14.5	1.1	0	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	10.3	6.8	2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.5	10.3	6.9	1.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	10.1	7	1.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	10	7	2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	9.9	7	1.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	9.8	7.1	2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	9.8	7.1	1.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	9.8	7.1	1.9	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.8	1.1	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.8	1.1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.8	1	0.5	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	10	11.1	16.4	53.3	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.3	11.7	11.2	14.5	46.6	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.9	9.8	16.4	53.4	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15	12.4	9.4	14	48.5	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.9	10.4	11.3	14.9	54.8	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.6	12.1	10.4	13.1	48.1	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	7.3	11.3	10.3	15.1	55.3	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	14.5	12.8	10.6	13.1	48.3	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	4	4.8	5.8	11.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	4	4.8	5.6	11.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.9	4.8	5.7	11.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.9	4.8	5.6	11.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.8	4.8	5.7	11.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.9	4.7	5.6	12	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.9	4.7	5.6	11.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.9	4.7	5.6	12	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	54.7	15.8	1.1	0.1	8.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.4	55.2	13	0.7	0	8.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	53.8	16	1.2	0	8.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.2	53.2	14.2	0.8	0	8.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	53.7	16.3	1.2	0.1	8.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.9	52.4	13.2	0.9	0	8.6
U2 - Efficacy additive alpha prior Beta(1, 1)	20.9	53.1	16.1	1.3	0.1	8.5
U3 - Efficacy additive alpha prior Beta(1, 1)	25	51.9	13.2	1.4	0	8.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	10.6	6.7	1.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	10.6	6.6	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	10.6	6.6	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	10.6	6.6	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	10.4	6.7	1.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	10.5	6.6	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.8	10.4	6.7	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	10.4	6.7	1.7	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.6	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.2	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1.2	0.6	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.3	48.2	27	8.3	0.9	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.8	49	22.7	6.7	0.5	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	45.5	27.9	8.7	0.7	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	48.7	22.6	6.5	0.3	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	45	28.7	8.2	0.9	4.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	46.5	22.6	6.1	0.3	4.3
U2 - Efficacy additive alpha prior Beta(1, 1)	12.5	44.2	29	9.1	0.9	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	20.7	46	22.3	6.2	0.5	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	9.1	8.4	4.2	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	9.1	8.4	4.1	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	9	8.4	4.2	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	9	8.4	4.2	1.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	8.9	8.6	4.1	1.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	8.8	8.6	4.2	1.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	6	8.8	8.6	4.2	1.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	6	8.9	8.5	4.2	1.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	1.8	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.7	1.7	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.7	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.7	1.8	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.7	1.8	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.7	1.7	1.4	0.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.9	1.3	0	0	0	39.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	58.4	1.3	0	0	0	40.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	59.1	1.1	0	0	0	39.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	59	1.1	0	0	0	39.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.7	1.3	0	0	0	40
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.6	1.1	0	0	0	40.3
U2 - Efficacy additive alpha prior Beta(1, 1)	58.8	1	0	0	0	40.2
U3 - Efficacy additive alpha prior Beta(1, 1)	59.5	0.8	0	0	0	39.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	1.9	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	1.9	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	1.9	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21	1.9	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	1.9	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21	1.9	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	21	1.9	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	21.1	1.9	0.5	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.3	59.5	10.4	1.1	0	4.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.1	56.4	8.6	1.1	0	4.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.6	57.7	11.5	1.4	0	4.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.7	54.9	9.4	1.3	0	4.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.8	57.6	11.5	1.3	0	4.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.2	54.6	9	1.4	0	4.8
U2 - Efficacy additive alpha prior Beta(1, 1)	25.9	55.8	12.3	1.3	0	4.7
U3 - Efficacy additive alpha prior Beta(1, 1)	29.9	54.9	8.9	1.6	0	4.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	12.7	5.9	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	12.6	6	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	12.6	6	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	12.6	6	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	12.5	6	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	12.6	6	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	12.4	6	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	12.5	6	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.2	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.1	1.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4	1.2	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.1	1.2	0.4	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	6.7	0	0	0	75.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	5.7	0	0	0	75.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.7	6.6	0	0	0	75.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	5.6	0	0	0	75.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18	6.5	0	0	0	75.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.7	5.9	0	0	0	75.4
U2 - Efficacy additive alpha prior Beta(1, 1)	18	6.3	0	0	0	75.7
U3 - Efficacy additive alpha prior Beta(1, 1)	18.6	5.9	0	0	0	75.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.3	2.3	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.4	2.3	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	2.3	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	2.3	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	2.3	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	2.3	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.4	2.3	0.4	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.3	2.3	0.4	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	1.4	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	1.4	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	1.4	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	1.4	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	1.9	6	48.1	42.8	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.3	2.6	7.6	49.4	38	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	1.2	5.5	45.7	46.4	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.5	2.6	7.5	47.6	39.7	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	1.4	5.4	45.6	46.4	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.7	2.8	6.8	47.1	40.5	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	1.1	5.7	45.1	46.9	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	2.6	2.8	6.9	47	40.6	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.2	3.8	8.7	11.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.2	3.2	3.8	8.7	11.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.2	3.8	8.4	11.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.2	3.1	3.8	8.5	11.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.1	3.8	8.3	11.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.2	3.2	3.8	8.3	11.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.8	8.3	11.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.2	3.8	8.2	11.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.7	31	28.8	4.8	0	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	38.3	31.9	24.7	4.4	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.6	31.2	29.9	4.7	0	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	41	30.4	23.7	4.2	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.6	29.3	30.1	4.4	0	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.1	29.4	24.2	3.6	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	34.7	29.5	30.4	4.7	0	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	43.1	28.9	23.9	3.5	0	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	8.5	8.9	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	8.7	8.9	2.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	8.5	9	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	8.4	8.9	2.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	8.2	9.1	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	8.2	9	2.4	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.8	8.2	9	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	8.2	9	2.4	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.2	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.1	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	1.4	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.7	30.1	24.3	4.3	0	10.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.4	30.4	21.7	4.1	0	10.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.2	29.7	26.4	4.1	0	10.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.4	28.8	22.4	3.8	0	10.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.6	29.1	25.8	4	0	10.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35	27.2	23	4.2	0	10.6
U2 - Efficacy additive alpha prior Beta(1, 1)	30.7	29	25.9	3.7	0	10.7
U3 - Efficacy additive alpha prior Beta(1, 1)	35.4	27.7	22.6	3.8	0	10.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	7.1	7.5	2.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	7.1	7.5	2.3	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	7.1	7.5	2.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	7	7.5	2.3	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	7	7.6	2.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	6.9	7.5	2.4	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.7	6.9	7.5	2.4	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.7	6.9	7.5	2.4	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.1	1.2	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.1	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.2	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.1	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.2	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.1	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.2	0.8	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.2	25.8	39.7	7.1	0.7	5.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.1	25.9	36.7	6.6	0.2	5.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	25.1	40.7	7.1	0.5	5.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	24.3	37.7	5.9	0.2	5.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	24.2	41.9	6.3	0.5	5.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.7	24	38.7	5.9	0.3	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	21.5	25.6	41.3	5.6	0.5	5.5
U3 - Efficacy additive alpha prior Beta(1, 1)	25.7	24.2	38.9	5.4	0.2	5.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	6.9	9.2	3.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	6.8	9.3	3.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	6.7	9.2	3.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	6.8	9.2	3.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	6.7	9.2	3.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	6.7	9.3	3.5	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.6	6.7	9.2	3.6	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.7	6.6	9.3	3.6	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.5	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.5	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.6	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.5	1	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.5	38.1	4.3	0	0	35.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.2	38.2	3.9	0	0	34.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	38.3	4.9	0	0	34.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.4	38	3.7	0	0	34.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	38.6	5	0	0	35
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23	37.9	4.1	0	0	35
U2 - Efficacy additive alpha prior Beta(1, 1)	21.7	38.3	5.1	0	0	34.9
U3 - Efficacy additive alpha prior Beta(1, 1)	23.7	37.3	4.2	0	0	34.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	8.3	3.5	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	8.4	3.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	8.3	3.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	8.3	3.6	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	8.2	3.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13	8.2	3.6	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.9	8.3	3.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.9	8.3	3.6	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.7	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.7	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.7	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	0.7	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.5	25.2	3.2	0.1	0	3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	71.7	22.6	2.8	0.2	0	2.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.3	23.8	2.9	0	0	3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	72.7	22	2.4	0	0	2.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.3	22.6	3.1	0.1	0	2.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	72.7	22	2.4	0.1	0	2.8
U2 - Efficacy additive alpha prior Beta(1, 1)	70.9	22.6	3.4	0.1	0	3
U3 - Efficacy additive alpha prior Beta(1, 1)	72.9	21.6	2.5	0.1	0	2.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	7.5	3.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	7.5	3.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.6	7.4	3.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	7.4	3.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.6	7.3	3.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.7	7.3	3.1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.7	7.2	3.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.7	7.2	3.1	0.4	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.7	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.3	16.2	14.9	36.4	11.4	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.8	17.8	13.9	33.3	8.4	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	15.6	15.3	37.9	11.6	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.2	17.4	13.9	31.7	9	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.3	16.1	14.6	36.6	11.6	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.8	16.8	12.8	33.5	8.3	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	18.5	15.8	15.6	37.7	11.6	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	26.5	18.1	13.2	31.7	9.7	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	4.8	6.1	8.1	5.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	4.9	6	8	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	4.8	6.1	8.1	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	4.7	6.2	8.1	5.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	4.7	6.1	8.1	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	4.7	6.1	8.1	5.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	4.7	6	8.2	5.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.3	4.7	6.1	8.1	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.6	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.8	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.8	1.6	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.8	1.6	2.3	0.9	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	34.7	28	24.1	6.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	39.3	26.4	20.3	4.4	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	32.5	29.5	24	6.2	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	38.1	27.7	20	4.3	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.6	30.1	31.5	24.2	6.5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	37	28.2	19.7	4.2	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	8.1	31.6	30.9	23.1	6.2	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	36.6	28.8	19.3	3.7	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.9	6.5	7.8	5.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.8	6.5	7.9	5.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	5.7	6.6	7.8	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	5.7	6.6	7.9	5.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	5.6	6.7	7.8	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	5.6	6.6	7.9	5.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.2	5.6	6.6	7.8	5.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	5.6	6.7	7.8	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	14.9	35.8	28.6	11.2	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.7	19	35	25.3	6.4	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	14.2	35.8	28.7	10.9	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.9	18.3	35.7	24	6.5	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	14.1	35.3	27	13	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	16.7	35.8	22.4	7	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	13	36.4	26.1	13.3	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	16.5	17.8	36.1	21.7	7.3	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.7	7.3	6.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.4	7.7	7.3	6.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.3	7.7	7.2	6.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.3	7.8	7.2	6.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.3	7.8	7.2	6.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.3	7.8	7.1	6.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	4.2	7.8	7.2	6.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4	4.3	7.7	7.2	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.9	2.6	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	6.5	13.4	42.7	30.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	8.8	14	41.2	24.7	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.3	6.6	13.8	42.7	29.5	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.3	7.2	13	42	25.4	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	6.8	13.7	42.1	30.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12	8.2	14	39.8	25.9	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	7	14.8	42.3	30	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	11.7	7.8	13.9	40.5	26	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	5.3	8.4	9.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.6	5.3	8.4	9.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.6	5.3	8.2	9.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.7	5.3	8.2	9.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.3	8.1	9.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.6	5.3	8.1	9.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.3	8.1	9.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.7	5.3	8.2	9.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.8	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.8	0.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15	14.2	48	16.3	4.5	2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	15	43.8	14.9	2.9	2.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.5	15.5	48.2	16.2	4.6	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23	14.8	42.5	14.9	2.8	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	14.4	50.2	17	4.5	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.7	15.1	43.4	14.7	3.1	2
U2 - Efficacy additive alpha prior Beta(1, 1)	13.6	13.8	49.6	16.2	4.7	2.1
U3 - Efficacy additive alpha prior Beta(1, 1)	21.3	16.2	41.9	15.5	3.1	2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.5	6.1	9.8	5.4	2.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.5	6.1	9.9	5.4	2.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	6	10.1	5.3	2.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.6	6	10	5.3	2.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	6.1	10.1	5.3	2.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	6.1	10.1	5.2	2.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.5	6.1	10.1	5.2	2.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.4	6.1	10	5.2	3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	1.9	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	2	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.3	2	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.3	1.9	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	2	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.3	2	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.3	2	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.3	1.9	1.6	0.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	40.7	33.8	11	2.6	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.7	48.7	25.3	8.6	1.6	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.8	40.8	32.8	10.7	2.8	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.7	44.6	27.7	8.2	1.7	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	38.7	32.8	11	3.5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.6	42.6	28.1	8.5	2.1	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	13.5	39	33.2	10.7	3.5	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	20.4	41.8	27.9	7.6	2.2	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	8.7	8.9	5.1	2.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	8.8	8.8	5.1	2.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	8.6	9.1	5	2.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	8.6	9	5.1	2.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	8.4	9.1	4.9	2.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	8.4	9.1	5	2.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	8.3	9.1	5.1	2.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.7	8.4	9.2	5	2.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.6	1.3	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.6	1.3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.6	1.3	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	44.5	12.8	5	1.1	16
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.7	41.2	10.5	3.9	0.9	15.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.3	42.9	13.8	4.9	1.2	15.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	41.7	10.8	4.3	0.8	15.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	44.5	12.6	5.3	1.2	15.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27	42	10.2	3.9	1	15.9
U2 - Efficacy additive alpha prior Beta(1, 1)	21.2	43.3	12.7	5.4	1.5	15.9
U3 - Efficacy additive alpha prior Beta(1, 1)	27.5	40.4	10.7	4.4	1.1	15.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	8.7	5.3	2.5	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	8.6	5.4	2.5	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	8.5	5.4	2.5	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	8.6	5.3	2.6	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	8.6	5.3	2.5	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	8.6	5.4	2.4	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	8.6	5.4	2.5	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	8.4	5.4	2.5	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.6	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.6	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1.1	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.1	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.1	0.8	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.9	54	11.2	2.2	0.3	7.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.9	51.5	8.4	1.7	0.2	7.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.8	53.8	11.8	1.8	0.3	7.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.8	49.5	8.6	1.3	0.2	7.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.1	53.3	11.2	1.9	0.2	7.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.5	48.1	9	1.7	0.2	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	26.1	52.5	11.7	2	0.2	7.5
U3 - Efficacy additive alpha prior Beta(1, 1)	32.8	48.9	9.1	1.4	0.2	7.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	10.9	6	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	10.9	5.9	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	10.8	6	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	10.8	5.9	2	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	10.7	5.9	2	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	10.7	5.9	2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	10.7	6	2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.3	10.6	6.1	2	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1.2	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1.2	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.1	1.2	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.1	1.2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.1	1.2	0.8	0.1	